6CCSACCM

Cooperation, Coordination and Multi-Agent Systems

Revision Notes

Academic Year 2025/26 — Semester 2

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These notes are structured to help you prepare for exams.
Each week includes: learning objectives, core concepts, theory, and worked examples.

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Module Overview

About This Module

This module focuses on **multi-agent systems**—systems consisting of multiple autonomous entities with distributed information, computational capabilities, and potentially conflicting objectives. These systems can be:

- **Cooperative**: sensor networks, teams of mobile robots in a warehouse
- **Competitive**: electronic marketplaces, resource/task allocation
- **Mixed**: collaborative platforms like Wikipedia or Stack Overflow

Agents may be entirely artificial, entirely human, or a mix of both.

Module code: 6CCSACCM **Semester:** 2 (AY 2025/26)

Module Aims

The module aims to:

- Discuss foundational and recent research on various aspects of **multi-agent systems**.
- Place particular emphasis on **coordination strategies** and **fostering cooperative behaviour**.
- Focus on **networks** and the **internet economy**.
- Introduce students to the field and help identify **potential research directions**.
- Develop skills in **critical analysis of research papers** and **formulating research questions**.

Learning Outcomes

By the end of this module, you will be able to:

Code	Learning Outcome
CCM1	Use **game-theoretic, algorithmic and computational tools** to model and analyse multi-agent strategic behaviour and its outcomes.
CCM2	Use **game-theoretic tools** to foster cooperation and stability in multi-agent systems.
CCM3	Use **algorithmic mechanism design techniques** to model and design incentives in multi-agent interaction.
CCM4	Apply **computational social choice techniques** to analyse multi-agent decision-making and improve the quality and stability of collective decisions.

Key Topics (Expected)

Based on the learning outcomes, this module is likely to cover:

Area	Topics
Game Theory	Strategic games, Nash equilibrium, coordination games, Prisoner's Dilemma, repeated games
Cooperation	Mechanisms for fostering cooperation, stability concepts, coalition formation
Mechanism Design	Incentive design, auctions, voting mechanisms, truthfulness
Social Choice	Voting rules, Arrow's theorem, manipulation, collective decision-making
Networks	Network effects, internet economy, platform design

Assessment and Feedback

Formative assessment opportunities:

- **Discussion exercises**: peer discussion and feedback
- **Tutorial questions**: regular problem sheets with feedback during sessions
- **Discussion forums**: post questions on KEATS
- **Office hours**: direct feedback opportunities

[TODO] Add assessment structure (exam format, coursework) when details are provided.

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1 Week 1: Non-Cooperative Games (Part I)

1.1 Topic Summary

Learning Objectives

By the end of this week, you will be able to:

1. Define the core concepts of non-cooperative game theory, including **strategy profiles**, **payoff functions**, and the distinction between cooperative and non-cooperative games.
2. Formulate real-world scenarios as **strategic-form (normal-form)** or extensive-form games involving rational, self-interested agents.
3. Identify and classify various types of strategies: **pure**, **mixed**, **dominant**, and **best response** strategies.
4. Analyse game-theoretic models to determine the existence and properties of **Nash equilibria** (in pure and mixed strategies).
5. Explain and interpret equilibrium behaviour in classical games: **Prisoner's Dilemma**, **Matching Pennies**, **Battle of the Sexes**, **Stag Hunt**, and coordination games.
6. Evaluate the potential for and limitations of non-cooperative solutions, including issues of **uniqueness**, **multiplicity**, and **inefficiency** of equilibria.
7. Demonstrate competency in using mathematical notation and reasoning to represent and solve strategic games.

The Big Picture

This week introduces **non-cooperative game theory**—the mathematical framework for analysing strategic interactions between rational decision-makers. The module sits at the intersection of two fields:

- **Algorithmic Game Theory (AGT):** Studies algorithms in strategic environments where inputs come from self-interested agents. Combines theoretical CS and economics to analyse equilibria and design efficient, fair mechanisms.
- **Computational Social Choice (COMSOC):** Studies collective decision-making and preference aggregation, asking how individual preferences can be combined into collective decisions.

The central question: *"What is a stable outcome when multiple self-interested agents interact?"*

What Examiners Like to Ask

- Define a game in **normal form**; write the formal tuple $\Gamma = (N, (S_i), (U_i))$.
- Given a payoff matrix, identify **dominated strategies** and solve by iterated elimination.
- Find all **pure strategy Nash equilibria** using best-response analysis.
- Compute **mixed strategy Nash equilibria** using the indifference principle.
- Compare **maximin** vs **minimax** strategies; apply the Minimax theorem to zero-sum games.
- Explain why Prisoner's Dilemma has a unique NE that is Pareto-inefficient.
- Distinguish **Nash equilibrium** from **strong equilibrium**.

1.2 Overview + Core Concepts + Extended Theory

Roadmap

1. Multi-agent systems and game types
2. Normal form games: formal definition
3. Dominance and iterated elimination
4. Nash equilibrium in pure strategies
5. Mixed strategies and expected payoffs
6. Nash equilibrium in mixed strategies
7. Computing mixed NE (indifference principle)
8. Maximin and minimax strategies
9. Zero-sum games and the Minimax theorem
10. Other solution concepts (strong equilibrium, correlated equilibrium)

1.2.1 Multi-Agent Systems

Multi-Agent System (MAS)

A **multi-agent system** consists of multiple autonomous entities (agents) with:

- Distributed information
- Computational capabilities
- Potentially conflicting objectives

Types:

Type	Examples
Cooperative	Sensor networks, robot teams in warehouses
Competitive	Electronic marketplaces, resource allocation
Mixed	Wikipedia, Stack Overflow

1.2.2 Normal Form Games

Non-Cooperative Game

In a **non-cooperative** (or competitive) game:

- Players take actions and enjoy outcomes *individually*
- There are **no binding contracts** among players
- Each player strategically optimises their own utility function

Normal Form Game (Formal Definition)

A game in **strategic (normal) form** is a tuple:

$$\Gamma = (N, (S_i)_{i \in N}, (U_i)_{i \in N})$$

where:

- N is a finite set of **players**
- S_i is the set of **strategies** for player $i \in N$
- $U_i : S \to \mathbb{R}$ is the **utility function** for player i, where $S = \times_{i \in N} S_i$ is the set of all strategy profiles

Strategy Profile Notation

A **profile** s is a vector of strategies $(s_1, \ldots, s_n)$ where $s_i \in S_i$.

Key notation:

- $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$ — profile without player i's strategy
- $s = (s_i, s_{-i})$ — decomposition of profile
- s_C — strategies of coalition $C \subseteq N$
- s_{-C} — strategies of players not in C

Exam Tip: Writing Game Definitions

When asked to "formulate as a game", always specify:

1. The set of players N
2. The strategy set S_i for each player
3. The utility function U_i (or payoff matrix for finite games)

1.2.3 The Prisoner's Dilemma

Prisoner's Dilemma

Setup: Two suspects are interrogated in separate rooms. Each can:

- **Cooperate (C):** Deny the crime
- **Defect (D):** Confess (rat out the other)

Payoff matrix:

	C	D
C	$-1, -1$	$-10, 0$
D	$0, -10$	$-5, -5$

Reading the matrix: First number = row player's payoff; second = column player's.

Key insight: The only stable outcome is (D, D), even though (C, C) gives both players a better payoff!

Prisoner's Dilemma: General Form

	C	D
C	R, R	S, T
D	T, S	P, P

Condition: $T > R > P > S$

- T = Temptation (defect while other cooperates)
- R = Reward (mutual cooperation)
- P = Punishment (mutual defection)
- S = Sucker's payoff (cooperate while other defects)

1.2.4 Dominance

Strict and Weak Dominance

For player i and strategies $s, t \in S_i$:

Strict dominance: s ***strictly dominates*** t if:

$$\forall s_{-i} \in S_{-i} : \quad U_i(s, s_{-i}) > U_i(t, s_{-i})$$

Weak dominance: s ***weakly dominates*** t if:

$$\forall s_{-i} \in S_{-i} : U_i(s, s_{-i}) \geq U_i(t, s_{-i}) \quad \text{and} \quad \exists s'_{-i} : U_i(s, s'_{-i}) > U_i(t, s'_{-i})$$

Dominant and Dominated Strategies

- Strategy s is **dominant** if it dominates *all* other strategies
- Strategy t is **dominated** if *some* other strategy dominates it

Key principle: Rational players never choose strictly dominated strategies.

Exam Tip: Dominance is Rare but Powerful

Dominant strategy is a very strong solution concept—no assumptions needed about other players. However, it rarely exists. In Prisoner's Dilemma, Defect is dominant for both players.

1.2.5 Iterated Elimination of Dominated Strategies (IEDS)

IEDS Algorithm

Procedure:

1. Find a strictly dominated strategy for any player
2. Eliminate it from the game
3. Repeat until no dominated strategies remain

Properties:

- **Strict dominance:** Order-independent (same final result)
- **Weak dominance:** Order-dependent (different elimination orders may give different results)
- Surviving strategies need not be dominant
- May end with multiple strategies per player

Example: IEDS on a 3×3 Game

Original game:

	L	C	R
T	$73, 25$	$57, 42$	$66, 32$
M	$80, 26$	$35, 12$	$32, 54$
B	$28, 27$	$63, 31$	$54, 29$

Step 1: For column player, R dominates L (compare: $32 > 25$, $54 > 26$, $29 > 27$). Eliminate L.

Step 2: For row player, T and B both dominate M. Eliminate M.

Step 3: For column player, C dominates R. Eliminate R.

Step 4: For row player, B dominates T ($63 > 57$). Eliminate T.

Solution: (B, C) with payoffs $(63, 31)$.

Verify NE: Row player: $63 > 54$ (B vs T against C). Column player: $31 > 29$ (C vs R against B). ✓

1.2.6 Nash Equilibrium (Pure Strategies)

Nash Equilibrium

A strategy profile $s^* = (s_1^*, \ldots, s_n^*)$ is a **(pure strategy) Nash equilibrium** if no player has a **unilateral improving move**:

$$\forall i \in N, \forall s_i' \in S_i : \quad U_i(s_i^*, s_{-i}^*) \geq U_i(s_i', s_{-i}^*)$$

Equivalently: each player plays a **best response** to the others' strategies.

Best Response

Strategy s_i is a **best response** to s_{-i} if:

$$s_i \in \arg\max_{s_i' \in S_i} U_i(s_i', s_{-i})$$

Exam Tip: Finding Pure Strategy NE

Method 1 (Best Response):

1. For each opponent strategy, find player's best response (underline/mark it)
2. NE = cells where *all* players are playing best responses

Method 2 (Check deviations):

1. For each cell, check if any player wants to deviate unilaterally
2. NE = cells with no profitable deviations

Example: Best Response Analysis

	L	C	R
T	$2, 2$	$1^*, 3^*$	$0^*, 1$
M	$3^*, 1^*$	$0, 0$	$0^*, 0$
B	$1, 0^*$	$0, 0^*$	$0^*, 0^*$

Reading: * marks best responses. NE = cells where both payoffs are marked.

Nash equilibria: None! No cell has both payoffs marked.

1.2.7 Coordination Games

Coordination Game

A game where players benefit from **choosing corresponding strategies**:

	Left	Right
Up	$1, 1$	$0, 0$
Down	$0, 0$	$1, 1$

Properties:

- **Two pure strategy NE:** (Up, Left) and (Down, Right)
- No dominant or dominated strategies
- Both NE are also **strong equilibria**

Battle of the Sexes

Players want to coordinate but have **different preferences**:

	Theater	Football
Theater	$2, 1$	$0, 0$
Football	$0, 0$	$1, 2$

Pure NE: (Theater, Theater) and (Football, Football)

Stag Hunt

	Stag	Hare
Stag	$4, 4$	$1, 3$
Hare	$3, 1$	$3, 3$

Story: Two hunters can cooperate for a stag (high reward) or safely catch a hare alone.

Pure NE: (Stag, Stag) and (Hare, Hare)—plus mixed equilibria (see below).

1.2.8 Mixed Strategies

Mixed Strategy

A **mixed strategy** is a probability distribution over pure strategies.

- $\Delta_i = \Delta(S_i)$ = set of probability distributions over S_i
- A pure strategy is a mixed strategy with probability 1 on one action
- Even with finite S_i, there are infinitely many mixed strategies

Expected Utility

When mixed strategy profile $\tilde{s} \in \Delta$ is played:

$$U_i(\tilde{s}) = \sum_{s \in S} U_i(s) \cdot \Pr(s|\tilde{s})$$

where:

$$\Pr(s|\tilde{s}) = \prod_{j \in N} \tilde{s}_j(s_j)$$

(product of individual probabilities, assuming independence)

Nash's Existence Theorem (1951)

Every finite game (finite players, finite action sets) has at least one **mixed strategy Nash equilibrium**.

1.2.9 Computing Mixed Strategy Nash Equilibria

Indifference Principle

In a mixed strategy NE where player i randomises over actions $A \subseteq S_i$:

Player i must be indifferent among all actions in A:

$$\forall a, a' \in A: \quad U_i(a, \tilde{s}_{-i}) = U_i(a', \tilde{s}_{-i})$$

Key insight: The *other* player's mixing probabilities make you indifferent!

Example: Matching Pennies

	H	T
H	$1, -1$	$-1, 1$
T	$-1, 1$	$1, -1$

No pure strategy NE (verify: every cell has a profitable deviation).

Mixed NE: Let Player 1 play H with probability p, Player 2 play H with probability q.

Player 2's indifference:

$$U_2(\text{H}) = U_2(\text{T}) \implies -p + (1-p) = p - (1-p) \implies p = \frac{1}{2}$$

By symmetry: $q = \frac{1}{2}$

Mixed NE: $\left(\frac{1}{2}, \frac{1}{2}\right), \left(\frac{1}{2}, \frac{1}{2}\right)$ with expected utility 0 for both.

Example: Stag Hunt Mixed NE

	Stag	Hare
Stag	$4, 4$	$1, 3$
Hare	$3, 1$	$3, 3$

Let $p = \Pr(\text{Player 1 plays Stag})$, $q = \Pr(\text{Player 2 plays Stag})$.

Player 2's indifference:

$$U_2(\text{Stag}) = U_2(\text{Hare}) \implies 4p + 1(1-p) = 3p + 3(1-p)$$

$$4p + 1 - p = 3p + 3 - 3p \implies 3p + 1 = 3 \implies p = \frac{2}{3}$$

By symmetry: $q = \frac{2}{3}$

Three NE total: Two pure + one mixed at $\left(\frac{2}{3}, \frac{2}{3}\right)$.

Exam Tip: Mixed NE Computation

1. Set up probability variables (p, q, etc.)
2. Write expected utility equations for each action
3. Set utilities equal (indifference condition)
4. Solve for the *other* player's probability
5. Verify the solution makes sense (probabilities in $[0, 1]$)

Common error: Solving for your own probability instead of the opponent's!

1.2.10 Maximin and Minimax Strategies

Maximin (Safety Level) Strategy

The **maximin strategy** for player i maximises their worst-case payoff:

$$s_i^{LS} \in \arg\max_{s_i} \min_{s_{-i}} U_i(s_i, s_{-i})$$

The **maximin value** (safety level) is:

$$U_i^{LS} = \max_{s_i} \min_{s_{-i}} U_i(s_i, s_{-i})$$

Intuition: If opponents play to hurt you as much as possible, what's your best guarantee?

Minimax Strategy (2-Player Games)

The **minimax strategy** for player i against player $-i$ minimises $-i$'s maximum payoff:

$$s_i^{MM} \in \arg\min_{s_i} \max_{s_{-i}} U_{-i}(s_i, s_{-i})$$

Intuition: How can you punish the opponent most, ignoring your own payoff?

Maximin in Prisoner's Dilemma

	C	D
C	$-1, -1$	$-10, 0$
D	$0, -10$	$-5, -5$

Row player's analysis:

- If commit to C: opponent plays D (minimises row), payoff $= -10$
- If commit to D: opponent plays D (minimises row), payoff $= -5$

Maximin: Choose D (guarantees $-5 > -10$)

Defect is both the dominant strategy AND the maximin strategy.

1.2.11 Zero-Sum Games and the Minimax Theorem

Constant-Sum and Zero-Sum Games

A two-player game is **constant-sum** if $\exists c$ such that:

$$\forall s \in S_1 \times S_2 : \quad U_1(s) + U_2(s) = c$$

For $c = 0$: **zero-sum game** (pure conflict—one player's gain is the other's loss).

Minimax Theorem (von Neumann, 1928)

For any finite, two-player, zero-sum game:

$$\max_{s_1} \min_{s_2} U_1(s_1, s_2) = \min_{s_2} \max_{s_1} U_1(s_1, s_2)$$

Consequences:

1. Each player's **maximin value = minimax value**
2. The common value is called the **value of the game**
3. **Maximin strategies = minimax strategies = NE strategies**
4. All Nash equilibria have the **same payoff vector**

Matching Pennies: Minimax Analysis

	H	T
H	$1, -1$	$-1, 1$
T	$-1, 1$	$1, -1$

This is a zero-sum game ($U_1 + U_2 = 0$ always).

Value of the game: 0

By Minimax theorem: The unique NE $\left(\frac{1}{2}, \frac{1}{2}\right)$ is also the maximin = minimax strategy for each player.

Computing Maximin/Minimax in General-Sum Games

For a general-sum 2-player game:

1. Player i's maximin depends *only* on i's utilities
2. Create a zero-sum game: keep i's utilities, negate them for $-i$
3. Solve the zero-sum game for NE
4. NE strategies = maximin strategies

1.2.12 Other Solution Concepts

Strong Equilibrium

A profile s is a **strong equilibrium** if no coalition $C \subseteq N$ has a joint improving move:

$$\nexists C \subseteq N, s'_C \in S_C : \quad \forall i \in C : U_i(s'_C, s_{-C}) > U_i(s)$$

Key facts:

- Every strong equilibrium is a Nash equilibrium (consider singleton coalitions)
- Much more rare than NE
- Prisoner's Dilemma has **no** strong equilibrium
- Coordination game: both pure NE are strong equilibria

Correlated Equilibrium

Players condition actions on a shared random signal from a trusted mediator.

- Every mixed NE induces a correlated equilibrium
- Set of correlated equilibria $\supseteq$ set of NE
- **Computationally easier:** Can be found via linear programming

ε-Nash Equilibrium

A profile where no player can gain more than $\varepsilon > 0$ by deviating:

$$\forall i, \forall s'_i : \quad U_i(s_i, s_{-i}) \geq U_i(s'_i, s_{-i}) - \varepsilon$$

Standard NE is the special case $\varepsilon = 0$.

Rationalizable Strategies

A strategy is **rationalizable** if it is a best response to some belief about opponents' strategies (consistent with common knowledge of rationality).

- Equivalent to surviving iterated elimination of never-best-responses
- Every NE strategy is rationalizable
- Set of rationalizable strategies $\supseteq$ NE strategies

1.2.13 Summary: Classic Games

Game	Pure NE	Mixed NE	Strong NE	Key Feature
Prisoner's Dilemma	1 (D,D)	Same	None	Dominant strategy, Pareto-inefficient
Coordination	2	+1	2	Multiple equilibria
Matching Pennies	0	1	0	Zero-sum, no pure NE
Battle of Sexes	2	+1	2	Asymmetric preferences
Stag Hunt	2	+1	2	Risk vs reward trade-off

1.2.14 Worked Examples from Exercises

Exercise: Dividing Money

Setup: Two players divide £10. Each names $x_i \in \{0, 1, \ldots, 10\}$.

- If $x_1 + x_2 \leq 10$: each gets their claim (remainder destroyed)
- If $x_1 + x_2 > 10$ and $x_1 \neq x_2$: lower claimant gets x_i, other gets rest
- If $x_1 + x_2 > 10$ and $x_1 = x_2$: each gets £5

Best responses:

Other's claim	Best responses
0	$\{10\}$
1	$\{9, 10\}$
2	$\{8, 9, 10\}$
$\vdots$	$\vdots$
5	$\{5, 6, 7, 8, 9, 10\}$
6	$\{5, 6\}$
7	$\{6\}$
8	$\{7\}$
9	$\{8\}$
10	$\{9\}$

Nash equilibria: $(5, 5)$, $(5, 6)$, $(6, 5)$, $(6, 6)$ — any pair where both are best responses to each other.

Exercise: Modified Stag Hunt

Modified payoffs:

	Stag	Hare
Stag	$4, 4$	$1, 3$
Hare	$3, 2$	$3, 3$

Whose mixing probability changes?

Player 2's utility changed ($1 \to 2$ in Hare-Stag cell).

Effect: Player 2's utility is used to determine Player 1's indifference condition.

Player 1's mixing probability changes: $p = \frac{1}{2}$ (was $\frac{2}{3}$).

Player 2's mixing probability stays: $q = \frac{2}{3}$ (Player 1's utilities unchanged).

1.2.15 Other Game Types (Preview)

Game Type	Description
Extensive form	Game tree; players move in turns (sequential)
Repeated games	Same game played multiple times; cumulative payoffs
Cooperative (coalitional)	Players form coalitions; binding contracts; characteristic function over coalitions

These will be covered in subsequent weeks.

2 Week 2: Non-Cooperative Games (Part II)

2.1 Topic Summary

Learning Objectives

By the end of this week, you will be able to:

1. Define the core concepts of non-cooperative game theory, including **strategy profiles**, **payoff functions**, and the distinction between cooperative and non-cooperative games.
2. Formulate real-world scenarios as **strategic-form (normal-form)** or extensive-form games involving rational, self-interested agents.
3. Identify and classify various types of strategies: **pure**, **mixed**, **dominant**, and **best response** strategies.
4. Analyse game-theoretic models to determine the existence and properties of **Nash equilibria** (in pure and mixed strategies) and other solution concepts.
5. Explain and interpret equilibrium behaviour in classical games such as the Prisoner's Dilemma, Matching Pennies, Battle of the Sexes, and coordination games.
6. Evaluate the potential for and limitations of non-cooperative solutions, including issues of **uniqueness**, **multiplicity**, and **inefficiency** of equilibria.
7. Demonstrate competency in using mathematical notation and reasoning to represent and solve strategic games.

The Big Picture

This week explores how strategic interactions evolve over time (**equilibrium dynamics**) and investigates the **efficiency of equilibria**. We study:

- **Congestion games:** A structured class of large games with concise representation
- **Best-response dynamics:** How agents naturally converge to equilibria
- **Potential functions:** A powerful tool for proving equilibrium existence
- **Price of Anarchy/Stability:** Quantifying the cost of selfish behaviour

The central question: *"How inefficient are equilibria compared to socially optimal outcomes?"*

What Examiners Like to Ask

- Define a **congestion game** formally; specify players, resources, actions, and cost functions.
- Given a network graph, compute player costs in different states and identify **Nash equilibria**.
- Explain why best-response dynamics converge in congestion games using **Rosenthal's potential function**.
- Calculate the **Price of Anarchy** and **Price of Stability** for given games.
- Prove bounds relating social cost to potential function values.
- Design games achieving specific PoA/PoS values (e.g., cut games with PoA = 2).

2.2 Overview + Core Concepts + Extended Theory

Roadmap

1. Large games and representation complexity
2. Congestion games: formal definition
3. Network/routing congestion games
4. Applications of congestion games
5. Best-response dynamics
6. Rosenthal's potential function
7. Potential games and the finite improvement property
8. Price of Anarchy and Price of Stability
9. Linear congestion games and tight bounds

2.2.1 Large Games and Representation

Representation Complexity

To represent an n-player game where each player has k actions in normal form, we need $\mathbf{k^n}$ numbers just to encode the utility functions.

Problem: For even moderately large k and n, this is intractable.

Solution: Study games with **concise structure** that allows compact representation and efficient reasoning.

Exam Tip

When asked about scalability of game representations, emphasize that normal form requires exponential space in the number of players, motivating structured game classes like congestion games.

2.2.2 Congestion Games

Congestion Game (Formal Definition)

A **congestion game** is defined by the tuple $(N, R, (A_i)_{i \in N}, (d_j)_{j \in R})$ where:

- $N = \{1, 2, \ldots, n\}$ is the set of **players**
- $R = \{1, 2, \ldots, m\}$ is the set of **facilities** (or resources)
- For each player i, $A_i \subseteq 2^R$ is the set of **actions**, where each action $a_i \in A_i$ is a subset of facilities
- For each facility $j \in R$, $d_j : \{0, 1, \ldots, n\} \to \mathbb{R}_{\geq 0}$ is the **cost (delay) function**

Key insight: The cost of a facility depends *only* on how many players use it, not on which specific players.

Player Costs in Congestion Games

Given action profile $a = (a_1, \ldots, a_n)$:

Congestion on facility j:

$$n_j(a) = |\{i : j \in a_i\}|$$

(number of players using facility j)

Cost to player i:

$$c_i(a) = \sum_{j \in a_i} d_j(n_j(a))$$

(sum of delay costs over all facilities in player i's action)

Exam Tip: Congestion Games vs Normal Form

Congestion games have **polynomial representation**: $O(n \cdot |R| + m \cdot n)$ to specify all action sets and delay functions. Compare to $O(k^n)$ for general normal form!

2.2.3 Network Congestion (Routing) Games

Network Congestion Game

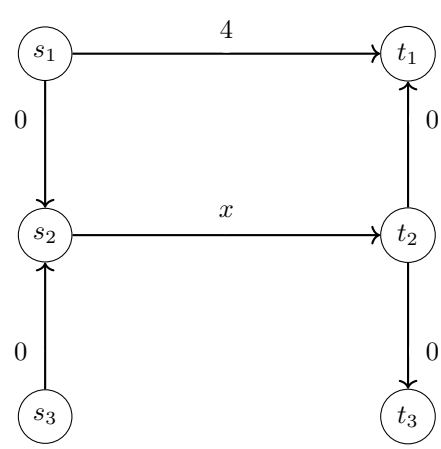
A **network congestion game** (or **routing game**) is a congestion game where:

- There is a directed graph $G = (V, E)$
- The resource set R corresponds to the edge set E
- Each player i has a source-destination pair (s_i, t_i)
- Player i's action set A_i = set of all directed paths from s_i to t_i

Interpretation: Each player chooses a route through the network; edge costs depend on congestion.

Example: Three-Player Routing Game

Consider the following network with three players:



Setup:

- Player 1: $s_1 \to t_1$ (two paths available)
- Player 2: $s_2 \to t_2$ (one path: direct)
- Player 3: $s_3 \to t_3$ (one path: via s_2, t_2)

Edge costs: Most edges have constant cost; edge (s_2, t_2) has cost $d(x) = x$ (identity function).

Player 1's strategies:

1. Direct path: $\{(s_1, t_1)\}$ with constant cost 4

2. Indirect path: $\{(s_1, s_2), (s_2, t_2), (t_2, t_1)\}$ with cost $0 + x + 0 = x$

State A (Player 1 takes direct route):

- Congestion on (s_2, t_2): 2 (Players 2 and 3)
- $c_1(A) = 4, \quad c_2(A) = 2, \quad c_3(A) = 2$
- Social cost: $4 + 2 + 2 = 8$

State B (Player 1 takes indirect route):

- Congestion on (s_2, t_2): 3 (all players)
- $c_1(B) = 3, \quad c_2(B) = 3, \quad c_3(B) = 3$
- Social cost: $3 + 3 + 3 = 9$

Analysis:

- **Nash Equilibrium:** State B (Player 1 prefers cost 3 over cost 4)
- **Social Optimum:** State A (total cost 8 vs 9)
- Selfish behaviour leads to **inefficient** equilibrium!

2.2.4 Applications of Congestion Games

Congestion games model decentralised decision-making in shared, congestible environments:

Domain	**Application**
Traffic & Transportation	Drivers choose routes to minimise travel time; analysis of road networks, navigation apps, highway design
Telecommunications	Data packets select routes/channels; TCP/IP routing, spectrum allocation, cellular load balancing
Cloud Computing	Jobs compete for CPUs, memory, bandwidth in data centres
Robotics	Multiple robots share warehouse aisles, coordinate movement
Smart Grids	Consumers choose when to use electricity; demand response
IoT Networks	Devices compete for transmission time, battery, sensors
Shared Economy	Ride-hailing, bike-sharing: too many users at one location reduces utility
AI/ML Resources	Competing agents select computing resources; federated learning
Cybersecurity	Attackers/defenders allocate resources across network surfaces
Urban Planning	Citizens choose public resources (parks, hospitals); overuse degrades quality

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2.2.5 Best Response Dynamics

Best Response Dynamics

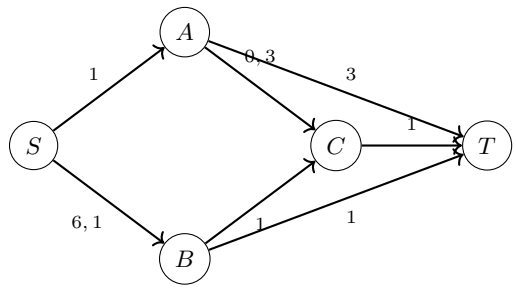
Algorithm:

1. Start with an arbitrary action profile $a = (a_1, \ldots, a_n)$
2. While $\exists$ player i who can decrease their cost by switching:
   - Player i switches to a **best response** (or any improving response)
3. If/when the process halts, return the current profile

Key observation: If best-response dynamics halts, the output is a **pure Nash equilibrium** (by definition: no player has a profitable deviation).

Example: Best Response Dynamics in a Routing Game

Consider two players (Red and Blue) travelling from S to T:



Edge labels show cost for 1 user, or (cost for 1, cost for 2) if congestion-dependent.

Initial state: Red takes upper path $S \to A \to T$, Blue takes lower path $S \to B \to T$.

- Red's cost: $1 + 0 + 3 = 4$
- Blue's cost: $6 + 1 = 7$

Round 1 (Red deviates): Red switches to $S \to A \to C \to T$ (cost: $1 + 0 + 1 = 2 < 4$).

Round 2 (Blue deviates): Blue switches to $S \to B \to C \to T$ (cost: $1 + 1 + 3 = 5 < 7$).

- Now both use edge (C, T), so Red's cost becomes $1 + 0 + 3 = 4$

Round 3 (Red deviates): Red switches back to $S \to A \to T$ (cost: $1 + 0 + 3 = 4$). Same cost, but let's say no strict improvement.

Equilibrium reached: Red on $S \to A \to T$ (cost 4), Blue on $S \to B \to C \to T$ (cost 3).

Neither player can improve $\Rightarrow$ **Nash Equilibrium**.

Question: Do Best Response Dynamics Always Converge?

No! Consider Matching Pennies:

	H	T
H	$1, -1$	$-1, 1$
T	$-1, 1$	$1, -1$

From any profile, some player has a profitable deviation $\Rightarrow$ **cycle forever**.

But: For **congestion games**, best-response dynamics *always* converges!

2.2.6 Rosenthal's Potential Function

Theorem (Rosenthal, 1973)

For every congestion game, every sequence of improvement steps is **finite**.

This property is called the **Finite Improvement Property (FIP)**.

Corollary

Every congestion game has at least one **pure strategy Nash equilibrium**.

Rosenthal's Potential Function

For any action profile a, define:

$$\boxed{\Phi(a) = \sum_{j=1}^{m} \sum_{k=1}^{n_j(a)} d_j(k)}$$

where $n_j(a)$ is the number of players using facility j under profile a.

Interpretation: Sum of "cumulative delay costs" over all facilities, summing delay values from 1 up to the actual congestion level.

Proof of Rosenthal's Theorem

Consider a single improvement step where player i switches from action a_i to b_i.

Change in player i's cost:

$$\Delta c_i = c_i(b_i, a_{-i}) - c_i(a_i, a_{-i}) = \sum_{j \in b_i \setminus a_i} d_j(n_j(a) + 1) - \sum_{j \in a_i \setminus b_i} d_j(n_j(a))$$

Since this is an improvement step: $\Delta c_i < 0$.

Change in potential:

$$\Delta\Phi = \Phi(b_i, a_{-i}) - \Phi(a_i, a_{-i}) = \sum_{j \in b_i \setminus a_i} d_j(n_j(a) + 1) - \sum_{j \in a_i \setminus b_i} d_j(n_j(a))$$

Key observation: $\Delta\Phi = \Delta c_i < 0$.

Since Φ takes finitely many values (finitely many profiles) and **strictly decreases** at each improvement step, the process must terminate. □

Exam Tip: The Potential Function Trick

The key insight is that the potential function exactly tracks individual cost changes:

- Facilities in $b_i \setminus a_i$: player joins, congestion increases by 1
- Facilities in $a_i \setminus b_i$: player leaves, congestion decreases by 1
- Facilities in $a_i \cap b_i$: no change to congestion

Both the player's cost change and the potential change depend only on the "boundary" facilities.

2.2.7 Potential Games

Exact Potential Game

A game $(N, (A_i)_{i \in N}, (U_i)_{i \in N})$ is an **exact potential game** if there exists a function $\Phi : A \to \mathbb{R}$ such that for every profile a, player i, and alternative action a_i':

$$U_i(a_i', a_{-i}) - U_i(a_i, a_{-i}) = \Phi(a_i', a_{-i}) - \Phi(a_i, a_{-i})$$

The function Φ is called the **exact potential function**.

Key insight: The change in any player's utility equals the change in the potential function.

Properties of Potential Games

- Every **exact potential game** has the Finite Improvement Property
- Every exact potential game has at least one **pure strategy Nash equilibrium**
- Local minima of Φ correspond to pure NE (for cost-minimisation games)
- **Congestion games are exact potential games** (Rosenthal's potential)

Exam Tip: Potential Games

To show a game is a potential game:

1. Propose a potential function Φ
2. Show that for any unilateral deviation, $\Delta U_i = \Delta \Phi$

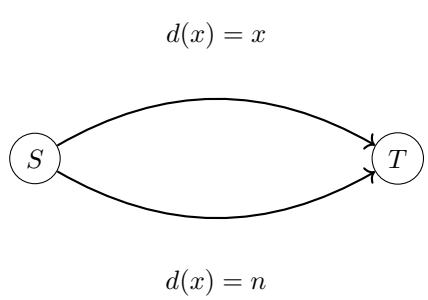
This immediately implies existence of pure NE and convergence of best-response dynamics.

2.2.8 Price of Anarchy and Price of Stability

Motivation: Inefficiency of Equilibria

Selfish behaviour can lead to socially suboptimal outcomes. We quantify this inefficiency by comparing equilibria to optimal solutions.

Pigou's Example



Setup: n players travel from S to T. Upper edge has cost = congestion; lower edge has constant cost n.

Selfish outcome: All players use the upper edge!

- If any player is on the lower edge (cost n), upper edge has $< n$ players, so cost $< n$
- Lower-edge player is "envious" and would switch
- **Equilibrium:** All on upper edge, each pays n, total cost $= n^2$

Social optimum: Split traffic equally

- $n/2$ on upper (each pays $n/2$), $n/2$ on lower (each pays n)
- Total cost $= \frac{n}{2} \cdot \frac{n}{2} + \frac{n}{2} \cdot n = \frac{n^2}{4} + \frac{n^2}{2} = \frac{3n^2}{4}$

Efficiency loss: $\dfrac{n^2}{3n^2/4} = \dfrac{4}{3}$

Price of Anarchy and Price of Stability

Given an optimisation objective (minimise cost or maximise welfare):

Price of Anarchy (PoA):

$$\text{PoA} = \frac{\text{value of worst Nash equilibrium}}{\text{value of optimal solution}}$$

Price of Stability (PoS):

$$\text{PoS} = \frac{\text{value of best Nash equilibrium}}{\text{value of optimal solution}}$$

Convention: For minimisation problems, PoA, PoS ≥ 1. For maximisation, invert the fraction so PoA, PoS ≥ 1.

Interpretation:

- PoA $= 1$: worst equilibrium is optimal (rare)
- PoS $= 1$: at least one optimal equilibrium exists
- Large PoA: selfish behaviour can be very costly

Prisoner's Dilemma: PoA and PoS

	C	D
C	$10, 10$	$0, 12$
D	$12, 0$	$1, 1$

Objective: Maximise social welfare (sum of utilities).

- **Unique NE:** (D, D) with welfare $1 + 1 = 2$
- **Social optimum:** (C, C) with welfare $10 + 10 = 20$

$$\text{PoA} = \text{PoS} = \frac{20}{2} = 10$$

Interpretation: Selfish behaviour loses a factor of 10 in efficiency!

Battle of the Sexes: PoA and PoS

	Theatre	Football
Theatre	$2, 1$	$0, 0$
Football	$0, 0$	$1, 2$

Objective: Maximise social welfare.

- **NE 1:** (Theatre, Theatre) with welfare $2 + 1 = 3$
- **NE 2:** (Football, Football) with welfare $1 + 2 = 4$
- **Social optimum:** welfare 4 (achieved by NE 2)

$$\text{PoS} = \frac{4}{4} = 1, \quad \text{PoA} = \frac{4}{3}$$

Interpretation: Best equilibrium is optimal, but coordination failure can cost $\frac{1}{3}$ of optimal welfare.

Three-Player Routing Game: PoA and PoS

Returning to our earlier example:

- **Unique NE:** State B (all use shared edge), social cost = 9
- **Social optimum:** State A, social cost = 8

$$\text{PoA} = \text{PoS} = \frac{9}{8}$$

2.2.9 Linear Congestion Games

Linear Congestion Game

A congestion game is **linear** if all cost functions have the form:

$$d_j(x) = a_j \cdot x + b_j$$

where $a_j \geq 0$ and $b_j \geq 0$ are constants.
Special case: If $b_j = 0$ for all j, the game has **identity cost functions**.

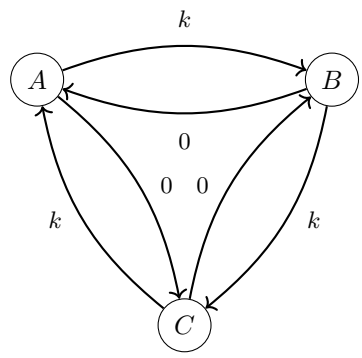
Theorem (Christodoulou and Koutsoupias, 2005)

For any congestion game with **linear cost functions**, the Price of Anarchy is at most **2.5**.

Theorem (Christodoulou and Koutsoupias, 2005)

There exist linear congestion games with 3 or more players where the Price of Anarchy equals **2.5** (tight bound).

Tight PoA = 2.5 Example (4 Players)



Players and destinations:

- Player 1: $A \to B$
- Player 2: $A \to C$
- Player 3: $B \to C$
- Player 4: $C \to B$

Edge costs: $d_j(x) = k \cdot x$ for "clockwise" edges; $d_j(x) = 0$ for "counter-clockwise" edges.

Optimal solution: Each player takes the direct path (cost k each).

- Total cost: $k + k + k + k = 4k$
- This is also a NE (any deviation uses at least one cost-k edge)

Worst NE: Players take longer paths:

- Player 1: $A \to C \to B$ (cost $0 + 2k = 2k$, but with congestion...)
- Player 2: $A \to B \to C$ (cost $2k + k$...)
- Player 3: $B \to A \to C$ (cost...)
- Player 4: $C \to A \to B$ (cost...)

After computing congestion effects:

- Players 1 and 2: cost $2k + k = 3k$ each (2 players on each edge they use)
- Players 3 and 4: cost $0 + 2k = 2k$ each
- Total: $3k + 3k + 2k + 2k = 10k$

Price of Anarchy: $\dfrac{10k}{4k} = \dfrac{10}{4} = 2.5$

2.2.10 Relating Potential Function to Social Cost

Lemma: Potential Bounds for Linear Congestion Games

For any linear congestion game and any strategy profile s:

$$\frac{1}{2} \cdot SC(s) \leq \Phi(s) \leq SC(s)$$

where $SC(s) = \sum_i c_i(s)$ is the social cost.

Proof Sketch

For linear costs $d_j(x) = a_j \cdot x + b_j$:

Social cost contribution from resource j:

$$SC_j = n_j(s) \cdot d_j(n_j(s)) = n_j(s) \cdot (a_j \cdot n_j(s) + b_j) = a_j \cdot n_j(s)^2 + b_j \cdot n_j(s)$$

Potential contribution from resource j:

$$\Phi_j = \sum_{k=1}^{n_j(s)} (a_j \cdot k + b_j) = a_j \cdot \frac{n_j(s)(n_j(s)+1)}{2} + b_j \cdot n_j(s)$$

Comparing:

- $\Phi_j \leq SC_j$ because $\frac{n(n+1)}{2} \leq n^2$ (equality when $n = 1$)
- $\Phi_j \geq \frac{1}{2} SC_j$ because $\frac{n(n+1)}{2} \geq \frac{n^2}{2}$

Exam Tip: Using Potential to Bound PoS

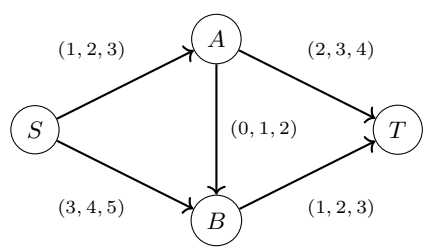
If a game has potential function Φ with $\alpha \cdot SC(s) \leq \Phi(s) \leq \beta \cdot SC(s)$, then:

$$\text{PoS} \leq \frac{\beta}{\alpha}$$

Because the best NE minimises Φ, and the optimal solution minimises SC.

2.2.11 Worked Examples from Exercises

Exercise 1: Finding NE in a Network Congestion Game



Setup: Three players, each travels $S \to T$. Edge labels (c_1, c_2, c_3) show cost when 1, 2, or 3 players use that edge.

Available paths:

1. $S \to A \to T$
2. $S \to B \to T$
3. $S \to A \to B \to T$

Solution approach:

1. List all strategy profiles (there are $3^3 = 27$, but symmetry helps)
2. For each profile, compute player costs
3. Check if any player can improve by switching

Candidate NE: All players take $S \to A \to T$.

- Edge (S, A): 3 players, cost = 3 each
- Edge (A, T): 3 players, cost = 4 each
- Total cost per player: $3 + 4 = 7$

Check deviations:

- Switch to $S \to B \to T$: cost $= 3 + 1 = 4 < 7$ ✓Profitable!

This is **not** a NE. Continue checking other profiles...

NE: (Found by systematic enumeration) Two players on $S \to A \to T$, one on $S \to B \to T$.

Exercise 2: Potential Function Bounds

Claim: For linear congestion games, $\frac{1}{2}SC(s) \leq \Phi(s) \leq SC(s)$.

Proof: For each resource j with n_j users and cost $d_j(x) = a_j x + b_j$:

Social cost contribution: $SC_j = n_j \cdot d_j(n_j) = a_j n_j^2 + b_j n_j$

Potential contribution:

$$\Phi_j = \sum_{k=1}^{n_j} (a_j k + b_j) = a_j \cdot \frac{n_j(n_j + 1)}{2} + b_j n_j$$

Upper bound: $\frac{n_j(n_j+1)}{2} \leq n_j^2$ (since $n_j + 1 \leq 2n_j$ for $n_j \geq 1$).

Lower bound: $\frac{n_j(n_j+1)}{2} = \frac{n_j^2}{2} + \frac{n_j}{2} \geq \frac{n_j^2}{2}$.

Summing over all resources gives the result. □

Exercise 3: PoS Bound Using Potential

Claim: If no resource is used by more than λ players, then PoS $\leq \lambda$.

Proof: Let s^* be the optimal profile and $\hat{s}$ be the best NE (minimises Φ).

Since NE minimises potential among all NE, and Φ strictly decreases along improvement paths:

$$\Phi(\hat{s}) \leq \Phi(s^*)$$

Using the bounds:

$$\frac{1}{\lambda} SC(\hat{s}) \leq \Phi(\hat{s}) \leq \Phi(s^*) \leq SC(s^*)$$

(The first inequality uses that with at most λ users, $\Phi_j \geq \frac{1}{\lambda} SC_j$.)

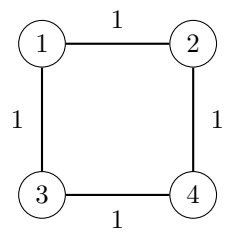
Therefore: $SC(\hat{s}) \leq \lambda \cdot SC(s^*)$, so PoS $\leq \lambda$. □

Exercise 4: Cut Games

Definition: Given graph $G = (V, E)$ with edge weights w_e:

- Each player i controls vertex v_i
- Strategy: place v_i in LEFT or RIGHT
- Utility: sum of weights of edges incident to v_i that cross the cut

(a) Design a cut game with PoA = 2:



Optimal: Diagonal cut (vertices 1,4 vs 2,3) cuts all 4 edges. Total utility = 8.

Worst NE: Horizontal cut (vertices 1,2 vs 3,4) cuts 2 edges. Total utility = 4.

PoA = $\frac{8}{4} = 2$.

(b) Prove PoA ≤ 2:

In any NE, each player must have at least half their adjacent edge weight in the cut (otherwise switching sides would be profitable).

For player i: $u_i \geq \frac{1}{2} \sum_{e \ni v_i} w_e$.

Summing: $\sum_i u_i \geq \frac{1}{2} \sum_i \sum_{e \ni v_i} w_e = \sum_e w_e$ (each edge counted twice).

Optimal welfare $\leq 2 \sum_e w_e$ (at most, each edge crosses the cut, counted for both endpoints).

Therefore: NE welfare $\geq \frac{1}{2}$ optimal $\Rightarrow$ PoA ≤ 2. □

2.2.12 Summary: Key Concepts

Concept	Key Points
Congestion Game	Players choose subsets of resources; cost depends on congestion only
Network Congestion Game	Resources are edges; actions are source-to-destination paths
Best Response Dynamics	Players iteratively improve; converges in congestion games
Rosenthal's Potential	$\Phi(a) = \sum_j \sum_{k=1}^{n_j(a)} d_j(k)$; decreases with each improvement
Finite Improvement Property	Every improvement sequence is finite $\Rightarrow$ pure NE exists
Price of Anarchy	Worst NE / Optimal (measures cost of selfishness)
Price of Stability	Best NE / Optimal (measures cost of decentralisation)
Linear Congestion Games	PoA ≤ 2.5; this bound is tight